Lowering the Barrier to Cyber Deception Education with Containerized Honeypot Labs

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Introduction

- What is cyber deception?
  - Cyber deception is one way to protect computers by creating fake services trick people who want to cause harm to a vulnerable system
- One of the primary cyber deception techniques are honeypots
- What is a honeypot?
  - Defensive decoy that simulates a computer system

Problem

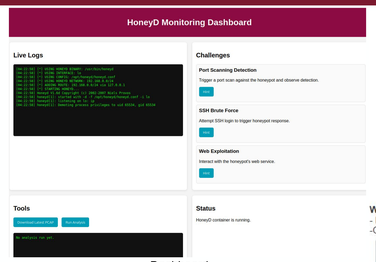
- Cyber deception education is not actively being taught or explored in classrooms
- Learning material is limited, costly, or hyperspecialized

Objective

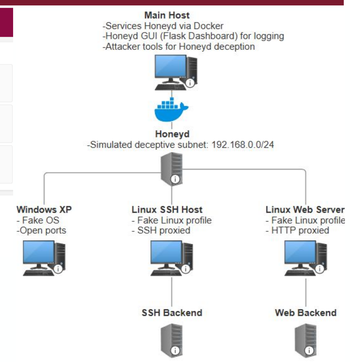
Develop a GitHub repository that helps students learn cyber deception techniques through a hands-on honeypot training (via Honeyd software) module featuring interactive challenges

Methods

- Using Docker, the user can create, access, and log the Honeyd subnet in one command
- From the user's machine, hacking tools are used to be detected by the honeypot
- Honeyd logs the user's activity and displays in Flask GUI Dashboard accessed on web browser



Dashboard



User Scenarios

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-21 17:21 MDT
Nmap scan report for 192.168.0.10
Host is up (0.019s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
80/tcp  open  http
135/tcp open  msrpc
445/tcp open  microsoft-ds
Device type: general purpose
Running: Microsoft Windows XP
OS CPE: cpe:/o:microsoft:windows_xp::professional
OS details: Microsoft Windows XP Professional
Network Distance: 1 hop

Nmap scan report for 192.168.0.20
Host is up (0.019s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
80/tcp open  http
Device type: general purpose|storage-misc
Running: Linux 2.4.X, Thecus embedded
OS CPE: cpe:/o:linux:linux_kernel:2.4.20 cpe:/h:thecus:4200 cpe:/h:thecus:n5500
OS details: Linux 2.4.20, Thecus 4200 or N5500 NAS device (Linux 2.6.33)
Network Distance: 1 hop

Nmap scan report for 192.168.0.30
Host is up (0.017s latency).
Not shown: 998 closed tcp ports (reset)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https
Device type: general purpose|media device
Running (JUST GUESSING): Linux 2.6.X|4.X (99%), Infomir embedded (89%)
OS CPE: cpe:/o:linux:linux_kernel:2.6.18 cpe:/o:linux:linux_kernel:4.3 cpe:/o:linux:linux_kernel:2.6 cpe:/h:infomir:mag-250
Aggressive OS guesses: Linux 2.6.18 (99%), Linux 2.6.9 - 2.6.27 (92%), Linux 4.3 (90%), Infomir MAG-250 set-top box (89%),
Linux 2.6.27 (89%), Linux 2.6.9 - 2.6.18 (89%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 3 IP addresses (3 hosts up) scanned in 5.00 seconds
```

Service/operating system scan on fake machines

Scenario 1

```
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-20 22:58:24
[DATA] max 4 tasks per 1 server, overall 4 tasks, 5 login tries (l:1/p:5), ~2 tries per task
[DATA] attacking ssh://192.168.0.20:22/
[22][ssh] host: 192.168.0.20   login: testuser   password: password123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-20 22:58:27
```

Command to brute force SSH credentials

```
*   Trying 192.168.0.30:80...
* Connected to 192.168.0.30 (192.168.0.30) port 80
> GET / HTTP/1.1
> Host: 192.168.0.30
> User-Agent: curl/8.5.0
> Accept: */*
>
* HTTP 1.0, assume close after body
< HTTP/1.0 200 OK
< Server: SimpleHTTP/0.6 Python/3.6.9
< Date: Tue, 21 Apr 2026 05:09:08 GMT
< Content-type: text/html; charset=ascii
< Content-Length: 436
<
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=ascii">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="entrypoint.sh">entrypoint.sh</a></li>
<li><a href="honeyd.conf">honeyd.conf</a></li>
<li><a href="scripts/">scripts/</a></li>
</ul>
<hr>
</body>
</html>
* Closing connection
```

Web enumeration through open port

Scenario 2

Live Logs

```
[23:21:18] honeyd[1]: Killing attempted connection: tcp (172.24.5.29:54308 -
192.168.0.20:1186)
[23:21:18] honeyd[1]: Killing attempted connection: tcp (172.24.5.29:54308 -
192.168.0.30:880)
[23:21:18] honeyd[1]: Killing attempted connection: tcp (172.24.5.29:54308 -
192.168.0.10:1119)
[23:21:18] honeyd[1]: Killing attempted connection: tcp (172.24.5.29:54308 -
192.168.0.20:3333)
[23:21:18] honeyd[1]: Killing attempted connection: tcp (172.24.5.29:54308 -
192.168.0.30:311)
[23:21:18] honeyd[1]: Killing attempted connection: tcp (172.24.5.29:54308 -
192.168.0.10:8083)
[23:21:18] honeyd[1]: Killing attempted connection: tcp (172.24.5.29:54308 -
192.168.0.20:3766)
[23:21:18] honeyd[1]: Killing attempted connection: tcp (172.24.5.29:54308 -
192.168.0.30:1045)
[23:21:18] honeyd[1]: Killing attempted connection: tcp (172.24.5.29:54308 -
```

Honeypot log output of outside scanning

Live Logs

```
[04:58:23] honeyd[1]: Killing attempted connection: tcp (192.168.0.20:22 -
172.24.43.114:56338)
[04:58:25] honeyd[1]: Connection established: (172.24.43.114:56338 - 192.168.0.20:22)
-> proxy to 127.0.0.1:2222
[04:58:25] honeyd[1]: Killing attempted connection: tcp (192.168.0.20:22 -
172.24.43.114:56340)
[04:58:25] honeyd[1]: Connection established: (172.24.43.114:56340 - 192.168.0.20:22)
-> proxy to 127.0.0.1:2222
[04:58:25] honeyd[1]: Killing attempted connection: tcp (192.168.0.20:22 -
172.24.43.114:56342)
[04:58:25] honeyd[1]: Connection established: (172.24.43.114:56342 - 192.168.0.20:22)
-> proxy to 127.0.0.1:2222
[04:58:25] honeyd[1]: Killing attempted connection: tcp (192.168.0.20:22 -
172.24.43.114:56362)
[04:58:25] honeyd[1]: Connection established: (172.24.43.114:56362 - 192.168.0.20:22)
-> proxy to 127.0.0.1:2222
[04:58:25] honeyd[1]: Killing unknown connection: tcp (192.168.0.20:22 -
```

Honeypot log output of SSH login

Scenario 3

Live Logs

```
[05:09:07] honeyd[1]: Connection request: tcp (172.24.7.15:56210 - 192.168.0.30:80)
[05:09:07] honeyd[1]: Killing attempted connection: tcp (192.168.0.30:80 -
172.24.7.15:56210)
[05:09:07] honeyd[1]: Connection established: (172.24.7.15:56210 - 192.168.0.30:80)
-> proxy to 127.0.0.1:8000
[05:09:08] honeyd[1]: Killing unknown connection: tcp (192.168.0.30:80 -
172.24.7.15:56210)
[05:09:08] honeyd[1]: Killing unknown connection: tcp (192.168.0.30:80 -
172.24.7.15:56210)
[05:09:08] honeyd[1]: Killing unknown connection: tcp (192.168.0.30:80 -
172.24.7.15:56210)
[05:09:08] honeyd[1]: Killing unknown connection: tcp (192.168.0.30:80 -
172.24.7.15:56210)
[05:09:08] honeyd[1]: Killing unknown connection: tcp (192.168.0.30:80 -
172.24.7.15:56210)
[05:09:08] honeyd[1]: Connection dropped by reset: tcp (172.24.7.15:56210 -
192.168.0.30:80)
```

Honeypot log output of website enumeration

Implications

- Accessibility
  - Allows both student and educator to create and deploy a fully containerized honeypot lab
- Approachability
  - Allows educators to teach from the provided curriculum in a safe and easy to deploy environment
- Academia
  - Allows students to learn from a structured curriculum of hands-on exercises, each with their own objective and learning outcomes

Conclusion

- Containerized Honeyd allows students to observe honeypots in action: attacker attraction, logging, and attack redirection
- Decoy targets demonstrate deception techniques and attacker monitoring
- Honeypots are a strong starting tool for cybersecurity education

Future Work

- User study to measure how effective the labs are at teaching cyber deception
- Measure usability, are they easy to deploy?
- Get IRB approval

Acknowledgements

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